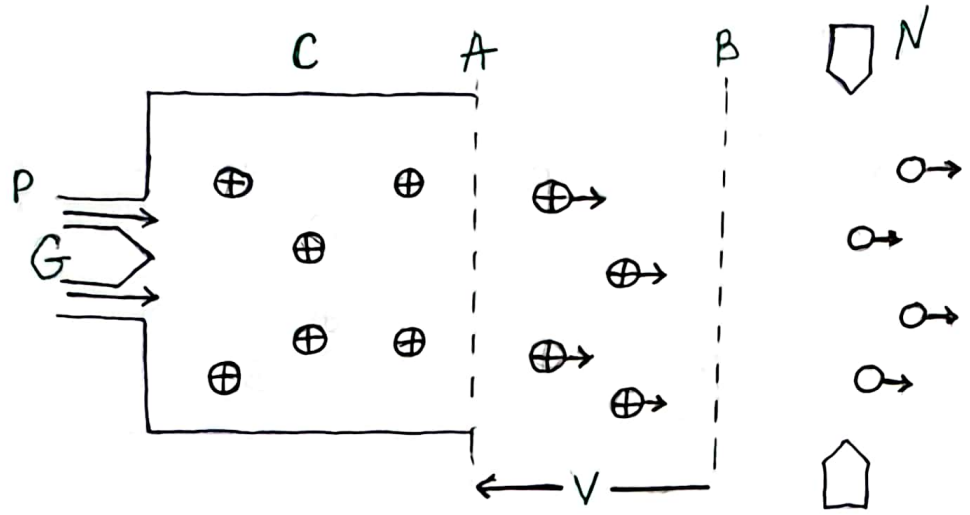
(4)



a). $F = \frac{dP}{dt}$

$F = \frac{N m v}{t} = R m v$ (1 ball)

$I = \frac{dq}{dt} = \frac{N z e}{t} = R z e$ (1 ball)

$\frac{F}{I} = \frac{m v}{z e}$ (1 ball)

$\frac{m v^2}{2} = V z e$ (1 ball)

$\Rightarrow v = \sqrt{\frac{2 z e V}{m}} \Rightarrow \frac{F}{I} = \sqrt{\frac{2 m V}{z e}}$ (1 ball)

b). $W = IV = R z e V$ (1 ball)

$\frac{F}{W} = \frac{R m v}{R z e V} = \sqrt{\frac{2m}{z e V}}$ (1 ball)

c). $\frac{F}{W} \sim \sqrt{m} \Rightarrow$ og'ir ionlar (1 ball)

$\frac{F}{W} \sim \sqrt{1/z} \Rightarrow$ bitta zaryadli ionlar. (1 ball)

$\frac{F}{W} \sim \sqrt{1/V} \Rightarrow$ kichik kuchlanish. (1 ball)

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